

Adaptive Quantization Method for CNN with Computational-Complexity-Aware Regularization

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Abstract—Quantization is a typical approach toward reducing processing time for inference of convolutional neural networks (CNNs). The key to reducing inference times without drastic decreases in accuracy is allocating optimal bit widths according to each layer or filter. One of the most promising ways to find optimal bit allocations is to learn quantization step size and weight parameters through gradient descent. The conventional method optimizes those parameters under the constraint for model size or memory footprint of CNN. However, the bit allocations obtained by the conventional method are not always optimal for inference time because the arithmetic intensity of CNN is significantly high, and the time spent on computing is a bottleneck for the inference time. In this paper, we propose a regularization method using a computational-complexity metric (which we call MACxbit) that is correlated with the inference time of quantized CNN models. The proposed method can obtain an optimal bit allocation that achieves better recognition accuracy under a specified computational-complexity target than the conventional method. For similar recognition accuracy on the optimized ResNet-18 model, the proposed method achieves 21.0% less inference time compared to the conventional method.

Index Terms—CNN, quantization, mixed-precision computing

I. INTRODUCTION

Convolutional neural networks (CNNs) have achieved high recognition performance in image classification tasks [1], [2]. CNN inference requires billions of multiply–accumulate (MAC) operations, causing a huge amount of latency and energy consumption. One effective approach to reducing latency and energy consumption is lowering computational costs by quantization of weight and activation values. For example, 4-bit fixed-point operations can be executed twice as fast as 8-bit operations on both commercial GPUs such as the NVIDIA Tesla T4 [3] and on dedicated accelerators [4], [5]. Recent works [6]–[12] have proposed methods (i.e., layer-wise or filter-wise quantization) for further reduction of computational costs without drastic accuracy reductions by allocating lower bit-widths to layers or filters that contribute less to recognition performance.

One of the most promising ways to find an optimal bit allocation is to employ an optimization method based on the gradient descent algorithm [13]–[15]. In gradient descent-based methods, the quantization step size which decides the bit width of weight is set as a learnable parameter (like the weight) and updated to minimize classification error through a stochastic gradient descent (SGD) algorithm. After iterative updates, we can derive an optimal bit allocation from the optimized quantization step size and weight. As one conventional

method, Ref. [15] proposes a regularization method based on model size (or memory footprint for CNN models). Using the regularization method, the quantization step size and weight are tuned to minimize classification error under a constraint for model size. In other words, the bit allocation can be optimized to achieve high recognition accuracy with specified model size. However, the conventional method focuses on only model size, and the optimized bit allocation is not always optimal for reducing inference time.

Regarding inference time of quantized CNN models, since the arithmetic intensity¹ of CNNs is significantly high, computation time (or computation speed) can be a bottleneck, rather than memory access time (or memory bandwidth) [17], [18]. Therefore, to reduce the inference time of quantized CNN models, computational complexity (e.g., the number of MAC operations) and computing performance (e.g., operations per second) should be considered rather than model size and memory footprint. Additionally, Refs. [4], [5] show hardware characteristics whose computation time is linearly proportional to bit precision. Considering the above, the inference time of a quantized CNN model can be estimated by accumulating products of the number of MAC operations and the bit precision (see Section III for details).

We propose a regularization method using a computational-complexity metric which is correlated with the inference time of quantized CNN models. With the proposed method, we can obtain an optimal bit allocation for achieving high recognition accuracy under specified computational complexity, or inference time. Moreover, we empirically show that models optimized by the proposed method achieve better accuracy with less inference time than did those by conventional method.

II. BACKGROUND

We explain the quantization procedure employed in this paper. As in Refs. [13], [14], when we quantize weight $\mathbf{W}$ by quantization step size s , the quantized weight $\mathbf{W}_{\text{int}}$ is expressed as

$$\mathbf{W}_{\text{int}} = \lfloor \mathbf{W}/s \rfloor, \quad (1)$$

where $\lfloor \cdot \rfloor$ is a rounding function to the nearest integer. In forward operations, we use the dequantized weight $\mathbf{W}_{\text{dq}}$ calculated as

$$\mathbf{W}_{\text{dq}} = \mathbf{W}_{\text{int}} \times s. \quad (2)$$

¹For instance, this metric is measured by the number of MAC operations per byte transferred from memory [16].